

MEASURING AI HARMS

March 2024

Adapting
Frameworks to
Assess AI's
Impact on
Civilians in
Conflict Zones

Written By:

Leo Podov Halbert
Mykhailo Hohol
Madeleine Northfield
Alexander Pymm
Mariam Seklaoui
Dawson Shinnars

LSE

Department of
**International
Development**

Background: This report was commissioned by the International Committee of the Red Cross, Regional Delegation to the UK and Ireland. It is in partial fulfilment of the MSc International Development and Humanitarian Emergencies at the London School of Economics and Political Science.

Acknowledgements: The consultancy team would like to thank everyone who contributed to this report's production. We are especially grateful to the ICRC and our liaisons there, Markus Geisser and Alexi Drew. Thank you for your time, support and vision. We would also like to thank our interviewees for taking the time to speak with us and their invaluable insights. Finally, we would like to thank our academic coach, Stuart Gordon, for his support throughout this project.

Limitations: The researchers acknowledge biases that may be present in this report due to limitations based on highly specialised knowledge of an emerging field, sample size, lack of gender diversity, representation of global south actors and discussion with affected populations. Nonetheless, interviews provided vital insight and validated the literature's themes.



Department of
**International
Development**



ICRC

Table of Contents

EXECUTIVE SUMMARY.....	4-5
ABBREVIATIONS.....	6
INTRODUCTION.....	7-10
Methodology.....	8
AI Applications & Civilian Harm Nexus.....	9-10
SECTION I: EXISTING METHODOLOGIES & FRAMEWORKS.....	11-16
Algorithmic Impact Assessment.....	11
Civilian Casualty Recording Methodologies.....	12
Collateral Damage Estimation Methodology.....	12
Conflict & Environment Observatory Guidelines	13
Cyber Attack Tracker.....	13
Data Protection Impact Assessment.....	14
Human Rights Impact Assessment	14
The Joint Targeting Cycle.....	15
Media Content Analysis.....	15-16
SECTION II: AI APPLICATIONS: RISKS & HARMS.....	17-19
AI-Autonomous Weapons Systems.....	17-18
AI-Decision Support Systems.....	18-19
AI-Cyber & Information Operations.....	19
SECTION III: ADAPTING FRAMEWORKS FOR AI.....	20-26
Civilian Casualty Recording	20-21
The Joint Targeting Cycle.....	21-23
Collateral Damage Estimation Methodology.....	23-24
Media Content Analysis	24-26
CONCLUSION & RECOMMENDATIONS.....	27-29
BIBLIOGRAPHY.....	30-36
APPENDICES	37-44
Appendix I: Final Terms of Reference.....	37
Appendix II: Original Terms of Reference.....	37-38
Appendix III: Information on Key Informant Interviews.....	38-39
Appendix IV: Interview Analysis.....	39
Appendix V: Supporting Images (Text Body).....	40-44

Abbreviation list

AI: Artificial Intelligence
AIA: Algorithmic Impact Assessment
AI-AWS: AI-Autonomous Weapons Systems
AI-DSS: AI-Decision Support Systems
AWS: Autonomous Weapons Systems
B'Tselem: Israeli Information Centre for Human Rights in the Occupied Territories
CCR: Civilian Casualty Recording
CCW: Convention on Certain Conventional Weapons
CDE: Collateral Damage Estimation
CEOBS: Conflict and Environment Observatory
CIVIC: Center for Civilians in Conflict
CSET: Center for Security and Emerging Technology
DDoS: Distributed Denial of Services attacks
DIHR: Danish Institute for Human Rights
DPIA: Data Protection Impact Assessment
DSS: Decision Support Systems
HRIA: Human Rights Impact Assessment
IBM: International Business Machines Corporation
ICRC: International Committee of the Red Cross
IDF: Israeli Defense Forces
IHL: International Humanitarian Law
iPRAW: International Panel on the Regulation of Autonomous Weapons
ISD: Institute for Strategic Dialogue
JTTP: Joint Tactics, Techniques, and Procedures for Intelligence Support to Targeting
JTC: Joint Targeting Cycle
LAWS: Lethal Autonomous Weapons
MCA: Media Content Analysis
MDH: Misinformation, Disinformation and Hate Speech
MHC: Meaningful Human Control
NATO: North Atlantic Treaty Organisation
NGO: Non-Governmental Organisation
OHCHR: United Nations High Commissioner for Human Rights
OSINT: Open-Source Intelligence
UK: United Kingdom
UN: United Nations
UNIDIR: United Nations Institute for Disarmament Research
US: United States

Introduction

For the past decade, a key strategic objective for the International Committee of the Red Cross (ICRC) has been “embracing the digital transformation,” with a focus on understanding “digital opportunities for accessibility and engagement” (ICRC, 2022, 13). Within the digital transformation sector, one of the fastest-growing opportunities and threats is the advancement of Artificial Intelligence (AI). The ICRC defines AI as the “use of computer systems to carry out tasks – often associated with human intelligence – that require cognition, planning, reasoning or learning” (ICRC, 2020a).

Over the past decade, developments in the use of AI in conflicts have been tracked and monitored by the ICRC and other international organisations, who view this technology as an opportunity and a challenge to the humanitarian sector (ibid.). The use of AI in conflict settings has also been of interest to the United Nations (UN), militaries and private technology corporations, such as Microsoft and International Business Machines Corporation (IBM). Despite this interest, interviewees unanimously highlight how much is unknown about AI on the battlefield and how it will impact civilians in conflict zones, one humanitarian lawyer compared it to computers in 1991, “we are just at the very beginning of all of this (Interviewee C, 2024).”

This project lays a foundation for understanding the impact of AI on civilian harm in conflict settings, specifically exploring existing frameworks for measuring harm and adaptations of those frameworks, where possible, to measure the harm of AI applications. Aiming to provide an entry point to the ICRC internally on these issues, this report will be guided by the following research questions:

1. What existing methodologies or frameworks exist for measuring harm from specific technologies or capabilities to civilians within conflict settings?
 - a. What are the strengths and weaknesses of these approaches?
2. Are any of these approaches capable of being adapted to the measuring and recording of harms caused by AI?
 - a. What types of harm might AI cause?
 - b. What kind of data points would be needed to be gathered to track these?

Methodology

This report is based on a two-stage research design that relies on desk-based research and online semi-structured interviews.

The desk-based research adhered to academic and grey literature, utilising scholarly articles, government and organisational reports, think-tank publications, and news media issues. It identified the current discourse surrounding AI's harm to civilians in conflict settings. The following focus areas were prioritised:

- Types of civilian harm
- Frameworks to assess civilian harm
- Applications of AI in conflict settings

The literature was analysed to identify applications of AI in conflict zones and their suitability to be assessed by harm measurement frameworks. The nine frameworks explored in Section I of this report vary in suitability but are intended to display the diversity and number of applicable harm measurement frameworks, addressing research question one. Sections II and III specifically address research question two in exploring the types of harm AI may cause and the applicability of these frameworks to measure civilian harm caused by AI in conflict settings.

Supplementary to the findings from the desk-based research, the report draws on insights from illustrative interviews. The following focus areas were prioritised:

- Technology providers and specialists
- Security sector and Ministry for Defense
- Humanitarian and human rights practitioners
- Researchers

See Appendix I and II for information on interviews and interview analysis.

The Nexus Between AI Applications and Civilian Harm

AI Harm: There are several challenges to defining AI harm. Perhaps most fundamental is establishing a clear causal link between a decision-making system, such as AI, and some resulting harm. As the Georgetown University Center for Security and Emerging Technology (CSET) discuss in their recent report, “The harm event or issue has to be directly linked to a consequence of an AI’s behaviour.” (Hoffmann and Frase, 2023, 18).

The CSET propose the following framework around which to incorporate specific elements of AI Harm:

1. Identify an entity that experienced
2. a harm event or harm issue (realised or anticipated) that
3. can be directly linked to a consequence of the behaviour of
4. an AI system

Taking civilians as our entity of interest, we explain how we will understand ‘civilian harm’ as an ‘event’ or ‘issue’.

Civilian Harm: There are a variety of accepted definitions of civilian harm within the international community. This report will utilise the definition and classification of civilian harm according to the “Protection of Civilians Allied Command Operations Handbook,” which identifies 20 harmful outcomes to assess future or realised civilian harm (NATO, 2020, 25).

The below table displays the complete list and their categorisation by ‘order of effect,’ referring to the immediacy of the harm to the civilian. Guidance from the ICRC and Civilians in Conflict (CIVIC) provide very similar examples in their approaches to assessing civilian harm (CIVIC, 2022; ICRC, 2023a). Our report proceeds with an understanding of civilian harm within this scope.

Order of Effect	Primary	Secondary	Tertiary
Type of Harm	• Death and injury to civilians • Sexual Violence • Destruction of civilian objects (i.e. houses) and critical infrastructure (i.e. water treatment plant)	• Forced displacement • Family separation • Inadequate access to food and water • Damaged infrastructure, affecting transportation routes, electricity, water and telecommunications access • Decreased mobility, lack of freedom of movement • Lack of access to medical attention • Damages to schools, disruptions to education • Disruption in financial services, access to banking and cash	• Weakened government • Weakened judicial services • Traumatized population • Sluggish and dysfunctional infrastructure • Lack of medical services • Market disruption, reduced economic activity • Cycles of violence • Increase in criminality • Spread of infectious diseases

(NATO, 2020, p. 25)

Linking AI Application to Harm: An understanding of how the behaviour of an AI application links to harm is required to explore the nexus between AI and civilian harm. The RAND Corporation constructed three categories of ‘AI Risk’ to describe the potentially harmful behaviours of AI applications.

Ethical	Operational	Strategic
Law of Armed Conflict Accountability and Moral Responsibility Human Dignity Human Rights and Privacy	Trust and Reliability Hacking, Data Poisoning, and Adversarial Attacks Accidents and Emergent Risks	Thresholds Escalation Management Proliferation Strategic Stability

(Morgan et al., 2020)

For each event or instance of civilian harm, anticipated or realised, the unique implications resulting from the behaviour of an AI system can be categorised by this taxonomy. This report will specifically explore the Ethical and Operational risks of AI applications, taking strategic-level implications to be beyond the scope of this report and noting that strategic risks are too broad for the scope of specific measurements of AI application's harm to civilians. While Operational risks identify instances where the behaviour of an AI system may contribute to tangible civilian harm, Ethical risks describe moral or legal implications that are typically less tangible but arise from aspects of AI behaviour that can cause civilian harm. By utilising the RAND AI risk taxonomy and its AI sub-risks by category, there is a clear understanding of how an event or instance of civilian harm can be linked to a consequence of an AI system's behaviour. Using this as a foundation, this report sets out to explore existing harm frameworks and their suitability to measure AI harm caused to civilians, as defined above.

SECTION III

ADAPTING FRAMEWORKS FOR AI

Section III provides means of adapting four of the frameworks found in Section I to measure the harm of AI applications, discussed in Section II. These four were selected based on measurement mechanisms that can be adapted to assess how AI applications harm civilians, including consideration of any relevant AI risks in line with the RAND taxonomy. Frameworks not yet viable for adaptation or already adapted to AI applications are highlighted in Section I.

Civilian Casualty Recording (CCR)

The CCR methodology measures the loss of civilian life in conflict zones, providing a valuable entry point for investigations into suspected breaches of IHL (OHCHR, 2019, 1). We now discuss some challenges that current CCR approaches face concerning capturing specific AI risks to civilians, such as ‘Accountability Gaps’ and ‘Black-Box Problems’.

Existing Challenges to CCRs: For CCRs to *directly* capture unpredictable and indiscriminate harm associated with Operational AI risks in conflict, such as the ‘Black-Box Problem,’ they must be capable of assessing the relevant combatants’ situational understanding of any AWS in use or being deployed (Cherry and Johnson, 2021; ICRC, 2018; iPRAW, 2018). Additionally, for CCRs to capture AI-created ‘Accountability Gaps,’ they must be capable of identifying and recording some degree of MHC over AWS deployed in conflict to ensure that any harmed civilians are not left “without a focal point of responsibility” (Morgan et al., 2020, 33).

UK-based NGO Article36 has suggested that MHC, and with it, sufficient information and situational understanding of AWS during the decision-making process, may be considered part of future investigations over the compliance of individual attacks with IHL (Article36, 2014).

Specific decision-making processes within the military chain of command exist in some approaches to CCR. However, where they are captured within CCR methodologies, as in OHCHR’s guidance (OHCHR, 2019), there are often practical challenges in gathering and verifying the data necessary to evaluate these elements. This is particularly evident when CCRs are conducted by third parties (such as NGOs, Peacekeeping Missions, or other third parties) and lack close and timely access to the chain of command. *Direct* measurement of MHC and situational awareness, and therefore Accountability Gaps and ‘Black-Box problems’, may be impossible in such situations.

Just as the identification of Unmanned Aerial Vehicles or automated air defence systems has helped hold states accountable in the face of imperfect information (Hofstra and Minor, 2014; Watts and Bode, 2021a), the recording of instances in which AWS are involved in civilian casualties may provide non-military actors with an indirect indicator, from which more detailed investigations into those decision-making processes can be conducted. An interviewee discussed that the ability to trace accountability is critical, noting that to hold groups accountable for their usage of AWS, we need to “know how to trace back before you put the technology out there” (Interviewee B, 2024). We discuss how CCRs may be adapted for this purpose.

Understanding Autonomy in AWS: For CCRs to be adapted to measure AWS's involvement in civilian casualties, one must first clearly define *autonomous* targeting systems. Consistent with the ICRC's definition of 'autonomy in critical functions' (ICRC, 2020b), lead researcher within the Global Security Initiative at Arizona State University, Heather Roff, proposes the following criteria by which LAWS may be evaluated:

- 1) "They can move independently through their environment to arbitrary locations;
- 2) They can select and fire upon targets in their environment; and
- 3) they can create and or modify their goals, incorporating observation of their environment and communication with other agents." - (Roff and Moyes, 2016a)

Roff classifies the autonomy of 284 different weapons systems across the five largest weapons-exporting countries. This definition is agreed upon in the international community, is critical to measuring LAWS-related harm, and is a first step in adapting CCR.

Identifying AWS in Conflict: There are also significant practical challenges to identifying AWS in conflict settings, even with an agreed-upon definition and understood capabilities. Older equipment, not previously understood as AWS, may be retrofitted with autonomous targeting systems (Watts and Bode, 2021a). Equipment fitted with autonomous systems is also not guaranteed to be employed at the moment of recording. A popular approach has therefore been to identify AWS in terms of their technical capabilities, determined from model specifications.

Researchers have employed novel open-source intelligence (OSINT) gathering methods, collating data from a variety of public and non-classified information sources, such as national defence organisations, NGOs and Defence Publications (Boulain and Verbruggen, 2017; Roff and Moyes, 2016b; Watts and Bode, 2021b; see Appendix V: Figures 5 and 6). However, Watts and Bode note some limitations of reliance upon open-source data. Without having physically observed the testing or development of the systems, they note the precise technical capabilities of equipment cannot be certain (2021b, 1). Nevertheless, existing approaches to standardising and cataloguing AWS may offer some hope to actors looking for a way to identify and document the use of AWS-enabled technologies in conflict settings. Once documented, these details may directly complement existing CCR approaches.

As explained previously, a more detailed evaluation of the military decision-making tools and the human-machine interaction may be required to directly capture 'Accountability Gaps' and 'Black Box Problems' posed by AWS to civilians in conflict. Evaluation of further Ethical Risks, such as 'Algorithmic Bias' from automated decision-making processes, or 'Threats to Privacy and Data Protection' (ICRC, 2020a), may also require data unavailable within typical CCRs.

The Joint Targeting Cycle (JTC)

The JTC can provide a framework for militaries to evaluate civilian harm within the broader decision-making contexts of their operations (Ekelhof, 2018a). Several members of the Group of Governmental Experts meeting on LAWS suggest the JTC framework might provide a basis for militaries to assess human control over LAWS in terms of the 'human-machine relation' between the AWS and its operator (ibid.; iPRAW, 2018; UK Government, 2018). These approaches may support efforts to identify 'Accountability Gaps' and 'Black-Box Problems' emerging from the use of AI to guide lethal weapons in conflict. We consider how both of these adaptations may be incorporated by militaries as part of the JTC.

Context-Dependent Human Control, Adapted for the JTC: The International Panel on the Regulation of Autonomous Weapons (iPRAW) suggest that rather than emphasise defining specific conceptions of autonomy in LAWS equipment, focus should be placed on the presence of human control of the operator over the weapons system throughout the execution of the JTC (iPRAW, 2018). This presents a framework for weapons systems that might be developed, one more flexible than those advocated for by the ICRC.

MHC during mission execution is defined by two analytical categories: ‘Control in Use’, and ‘Control by Design’, as defined in the below table:

	Situational Understanding	Intervention
Control by Design (Technical Control)	Ability to monitor information about environment and system	Modes of operation that allow human intervention and require them in specific steps of the targeting cycle
Control in Use (Operational Control)	Appropriate monitoring of the system and the operational environment	Authority and accountability of human operators, teammates and commanders; abide by IHL

(iPRAW, 2018, tbl. 1)

Within the context of an engagement, one can apply the iPRAW approach of human control through execution by breaking down Phase 5 of the JTC, ‘Mission planning and force execution’, into a 7-step, tactical-level cycle: ‘Find, Fix, Track, Target, Engage, Exploit, Assess’, or F2T2E2A (NATO, 2021, sec.5.2; see Appendix V: Figure 7). At each step, one can apply the criteria for both ‘Control in Use’ and ‘Control by Design’.

An operator, having found, fixed and tracked a given target, is capable at Step 4 of reflecting on any subsequent changes in the mission environment, consulting with the AWS to determine any system-level factors or technical challenges likely to prevent the intended outcome from being achieved, and ultimately maintaining the ability to abort or alter the course of engagement at any moment before Step 5.

iPRAW also offers several interpretations for how a ‘minimum requirement for control’ might be defined within the JTC (see Appendix V: Table 1). Additionally, JTCs could be adapted to measure AI-specific risks in a military engagement that may cause harm to civilians even with MHC. This might include identifying any ‘Accountability Gaps’ produced that would separate the operator as a decision-maker from the resulting civilian harm or evidence of ‘Black-Box Problems’ in an unpredictable or unreliable AI targeting system.

Testing and Verifying Human Control Framework: There are challenges to militaries implementing these adaptations into existing JTC frameworks. iPRAW and the International Committee for Robot Arms Control (ICRAC) suggest several specific instruments that might be used to validate and ‘test’ human control and understanding across the LAWS lifecycle and thus measure the Operational risks of AI (Gubrud and Altmann, 2013; iPRAW, 2019). Gubrud and Altmann highlight the potential use of “glass box” recording devices, capable of securely recording and reporting operator involvement and engagement “together with a time stamp of the moment the engagement command was issued.” (2013, 6).

Widespread adoption and implementation of these instruments by militaries at the deployment stage would be necessary to facilitate this adaptation of existing reporting processes within the JTC. As one respondent observed, however, government agencies and militaries currently lack the volume of technical expertise required to effectively implement 'self-policing' methodologies and the transparency to facilitate external assessments (Interviewee F, 2024). This would require significant policy changes, such as the implementation of sweeping and complex regulations through the Convention on Certain Conventional Weapons (CCW) (iPRAW, 2021). We discuss these possible steps as part of our recommendations.

Collateral Damage Estimation (CDE) Methodology

As cyberwarfare emerges as a theatre of modern conflict, there is extensive work across humanitarian, technological and military spheres incorporating CDE into the planning of cyberwarfare operations. After considering how a CDE methodology might be adapted to cyber attacks more generally, we consider some potential challenges in extending those predictions to AI-supported attacks in light of the unique predictability and accountability risks posed.

Challenges to Existing CDE Methodologies: The position taken by the ICRC and many states has been "that the employment of cyber capabilities in armed conflict must comply with all the principles and rules of IHL." (ICRC, 2015, 40). However, there is no standardised approach for capturing these impacts, so significant challenges remain to ensure compliance.

Efforts have focused on precisely how the rules of IHL and assessments of expected civilian harm can be translated from traditional kinetic contexts to cyberspace (Gisel and Olejnik, 2019). Conceptual challenges include identifying what it means for a cyber operation to constitute an "attack" and identifying the boundaries of which non-physical effects should be considered "harms" under IHL (Romanosky and Goldman, 2017). Whilst the Tallinn Manual has been identified as a potential basis for building a more substantial legal address to these questions (ICRC, 2021c), there is a lack of consensus. One senior figure from the humanitarian digital policy space made clear their feelings that there were not yet any "convincing frameworks" and that greater efforts were needed to "simply start to figure out what kind of incarnations it [digital harm] could have." (Interviewee H, 2024).

Adapting the CDE to Cyber: Setting aside these more fundamental questions, we consider some existing attempts to adapt the CDE for practical cyber use cases. Some approaches have maintained an understanding of cyber collateral damage that is exclusively limited to cyber as a means of warfare. For example, researchers at the Delft University of Technology have proposed classifying and estimating cyber operations' effects by using AI models to evaluate behaviour across real and simulated cases (Maathuis et al., 2021). Their results, a 'multi-layered fuzzy model' were presented to the Netherlands Ministry of Defence and evaluated by a team of military-technical experts to inform their assessment of likely damage to civilian objects by cyber operations. These results were then used to notify a proportionality assessment of cyber into the JTC framework, mimicking those done for kinetic operations. Several similar approaches have also been proposed (Tavana et al., 2014; Zheng et al., 2009).

Alternative approaches have defined cyber collateral damage not according to how they are produced (i.e. cyber attacks), but by the nature of the effects (cyber effects). In an adaptation proposed by RAND (see Appendix V: Figure 8), cyber collateral damage is defined as follows:

“Unintended harm to a computer or information system that is not the target of a lawful cyber operation. Where “harm” is defined as either a) the deletion, manipulation, or alteration of computer code governing the operation of hardware or software that is not specifically intended by the party conducting a lawfully-authorized operation, or b) the compromise of the integrity or availability of a computer network or data, or exfiltration of data, that is not specifically intended by the party conducting a lawfully-authorized operation.” (Romanosky and Goldman, 2017, 244–5).

The authors note that this has the potentially unintended consequence of capturing cyber harm that results from kinetic operations. They go on to note that this may be a strength in some instances and allow the adapted CDE to capture ‘civilian cyber harm’ in settings where the potential implications of an operation are less clear, such as the disruption to essential online services that resulted from the US’ physical destruction of Iraqi satellite and communication equipment in 2003 (ibid., 246).

Overall, it is estimated that CDEs adapted to cyberspace will face significant challenges to reliably estimating the civilian harm associated with cyber operations, particularly with regard to unpredictable secondary, or higher-order impacts (Romanosky and Goldman, 2016).

Applications of a Cyber CDE to AI Applications: Should a standardised CDE methodology be adapted to cyber operations, there are likely to be unique challenges to measuring AI-assisted methods of warfare. CDE methodologies for non-AI-assisted methods of attack are generally informed by previous weapons testing review processes, predictable data for future attacks, however (ICRC and ARES, 2016), AI-supported methods may be capable of changing their initial properties, making predictions very challenging.

In the extreme, some methods of AI-supported cyberwarfare may be effectively impossible to predict or control by those who deploy them. Once deployed, autonomous capabilities, such as AI-supported ‘cryptoworms’, may be able to embed malicious code before intelligently targeting and self-propagating “from one system to another by itself, with no human intervention” (Zouave et al., 2020, 10). In such instances, the scope for disruption, or ‘cyber harm’, resulting from the interaction of sophisticated AI attack methods and a network of complex systems is likely incalculable given current methods. Beyond ‘Black-Box Problems’, the possibility of ‘Emergent Risks’ presented by AI-assisted cyber attacks may present an even more significant challenge to current efforts to adapt CDEs to cyberspace.

In order for CDEs to effectively measure AI-supported attacks, conceptual issues underlying the adaptation of CDEs to *cyberspace* need to be addressed first. Despite the progress of initiatives such as the Tallinn Manual, consensus and formal legal instruments are needed to determine the perimeters of incidental cyber harm and how collateral damage should be considered.

Media Content Analysis (MCA)

MCA has been widely adopted to measure instances of MDH across various conflict settings and media platforms, as discussed in Section 1. Generative AI eases the ability of actors to create misleading text, image and video content, posing a particular challenge to misinformation on social media platforms. Therefore, a focus on new and emerging approaches to identify instances of Generative AI is required to identify its use in producing mis/disinformation. As Generative AI technologies continue to grow more sophisticated, the authenticity of misleading content that can be generated and the scale at which it can be produced is only likely to increase further on social media platforms (Marcellino et al., 2023).

The ability of current approaches within MCA to keep pace with the rate of technological development is unlikely, due to being conducted by individual human assessment and more traditional software tools.

Measuring Generative AI Disinformation through MCA: A recent report by the Brookings Institute (Srinivasan, 2024; see Appendix V: Table 2) identifies four potential approaches to identifying the use of Generative AI in content created and distributed online:

- **Watermarking:** A process by which the developer of a given Generative AI model may embed a ‘watermark’ within the content generated by the model. This may later be detected by ‘watermark detection’ software on a platform.
- **Post-hoc detection software:** A broad term to describe AI models designed to analyse published content for evidence of Generative AI used in its creation.
- **Retrieval-based detection:** Content generated by Generative AI models may be stored within a database maintained by the model’s developer. Publication of content could then be checked against a database of known instances of generated content.
- **Content provenance:** A process by which the Generative AI model embeds information around the content’s provenance within the item’s metadata for later identification by detection software.

Without effective policies requiring the cooperation of model developers, ‘watermarking’, ‘retrieval-based’ and ‘content provenance’ approaches will likely remain impractical for evaluations of MDH harm caused by Generative AI.

Whilst post-hoc detection methods may have greater practicality in supporting the evaluation of MDH harm, significant technical challenges remain and continue to grow. Primarily, the gap between the production of content and AI detection must be narrowed before they can be expected to produce reliable results.

Currently, Meta has several post-hoc detection tools in development. According to a press material released by the company in 2020, their SimSearchNet++ technology promises “to match near-duplications of known misinformation at scale” (Meta, 2020).

The voluntary pledge of 20 leading tech companies, including OpenAI, Google, Meta, Microsoft, Amazon, and TikTok, to collaborate on countering misleading “deepfake” content during the 2024 Munich Security Conference has been taken as a positive sign (Hodgson, 2024), although more concrete steps are needed. Organisations are expected to deploy techniques such as ‘watermarks’ to their platforms in the coming months (Herrero et al., 2024). As argued by a research analyst, voluntary initiatives are unlikely to address the challenges posed by “deepfakes” on a larger scale (Interviewee D, 2024).

A probable challenge with regulation remains in ratifying current legislation. The European Union’s AI Act proposed in 2021 mandates clear labelling for all AI-generated content, while the Digital Services Act aims to compel the tech industry to combat deepfakes. Still, it has not yet been enacted (Future of Life Institute, 2024).

Retrieval-based detection appears to be an optimal choice for indicating content generated by AI, however, it requires vast data storage, which entails privacy and cost questions. As an interviewee mentioned, it's unlikely that companies will either absorb the expenses themselves, given their profit-driven nature, or government subsidies for humanitarian purposes would be considered, considering the sector's limited resources (Interviewee D, 2024).

The primary objective of this framework's adaptation is to integrate new technologies and collaborate with stakeholders across various sectors to stay ahead of emerging challenges posed by AI. Speaking on the implications of AI-driven mis/disinformation campaigns for humanitarian organisations at an operational level, an interviewed senior humanitarian practitioner from a major humanitarian agency with a particular focus on technology, noted that "Humanitarians had long operated in environments of information warfare. However, the introduction of AI contributed to lowering the threshold of use and new levels of scalability of such campaigns. Ultimately, AI-driven information warfare risked contributing to the collapse of notions of truth and fact." The interviewee concluded that "If humanitarians do not manage to have a minimum of trust and facts, it will be challenging to have principled action, and I see that as a real threat" (Interviewee G, 2024)." MCA adaptations will require using the above approaches and other measuring tools employed by the states and private companies.



Bibliography

- Amnesty International, 2023. A Death Sentence For My Father: Meta's Contribution to Human Rights Abuses in Northern Ethiopia. Amnesty International, London. Available at: <https://www.amnesty.org.uk/files/2023-10/Amnesty%20International%20Tigray%20Meta%20Report.pdf?VersionId=TcqfAYDhZAX2MhsXMj7acym5gc6GMTQG>. (accessed 21.2.24).
- Amnesty International, 2022. The social atrocity: Meta and the right to remedy for the Rohingya. Amnesty International, London. Available at: <https://www.amnesty.org/en/documents/asa16/5933/2022/en/>. (accessed 14.2.24).
- Article36, 2014. Key areas for debate on autonomous weapons systems (Briefing Paper). Article36. Available at: <https://www.article36.org/wp-content/uploads/2014/05/A36-CCW-May-2014.pdf>. (accessed 8.2.24).
- Bagshaw, S., 2022. Committing to civilian casualty tracking in the future political declaration on the use of explosive weapons in populated areas (Policy Briefing). Article36. Available at: <https://article36.org/wp-content/uploads/2022/04/Article-36-Casualty-Tracking-and-the-Political-Declaration-on-EWIPA.pdf>. (accessed 6.2.24).
- Boulanin, V., Verbruggen, M., 2017. Mapping the Development of Autonomy in Weapon Systems. SIPRI. Available at: <https://www.sipri.org/publications/2017/policy-reports/mapping-development-autonomy-weapon-systems>. (accessed 12.2.24).
- B'Tselem, 2024. B'Tselem - The Israeli Information Center for Human Rights in the Occupied Territories [WWW Document]. B'Tselem.org. URL <https://www.btselem.org/> (accessed 8.3.24).
- Buolamwini, J., Gebru, T., 2018. Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification, in: Proceedings of the 1st Conference on Fairness, Accountability and Transparency. Presented at the Conference on Fairness, Accountability and Transparency, PMLR, pp. 77–91.
- CEOBS, 2022. About the Conflict and Environment Observatory.
- CEOBS, ZEN, 2024. Ukraine conflict environmental briefing: Nature. CEOBS. URL <https://ceobs.org/ukraine-conflict-environmental-briefing-nature/> (accessed 10.3.24).
- Cherry, J., Johnson, D., 2021. Maintaining Command and Control of Lethal Autonomous Weapon Systems: Legal and Policy Considerations. Southwest. J. Int. Law 27, 1–27.
- CIVIC, 2022. A Primer on Civilian Harm Mitigation in Urban Operations. Center for Civilians in Conflict. Available at: https://civiliansinconflict.org/wp-content/uploads/dlm_uploads/2022/11/CIVIC_UrbanWar_FINAL-2022_Web-1.pdf. (accessed 24.2.24).
- Columbia Law School, CIVIC, 2012. The Civilian Impact of Drones: Unexamined Costs, Unanswered Questions, Modern Issues in Conflict. Available at: <https://civiliansinconflict.org/publications/research/civilian-impact-drones-unexamined-costs-unanswered-questions/>. (accessed 25.2.24).
- Coppi, G., Moreno Jimenez, R., Kyriazi, S., 2021. Explicability of humanitarian AI: a matter of principles. J. Int. Humanit. Action 6, 19. <https://doi.org/10.1186/s41018-021-00096-6>
- CyberPeace Institute, 2023. Frequently Asked Questions: Data & Methodology | CyberPeace Institute [WWW Document]. URL <https://cyberconflicts.cyberpeaceinstitute.org/faq/data-and-methodology> (accessed 8.3.24).
- CyberPeace Institute, 2022a. Cyber Dimensions of the Armed Conflict in Ukraine (Quarterly Analysis Report Q3 July to September 2022) (Quarterly Analysis). Available at: https://cyberpeaceinstitute.org/wp-content/uploads/Cyber%20Dimensions_Ukraine%20Q3%20Report.pdf#page=13.

CyberPeace Institute, 2022b. Case Study: Viasat Attack | CyberPeace Institute [WWW Document]. URL <https://cyberconflicts.cyberpeaceinstitute.org/law-and-policy/cases/viasat> (accessed 10.3.24).

Davies, H., McKernan, B., Sabbagh, D., 2023. 'The Gospel': how Israel uses AI to select bombing targets in Gaza. The Guardian. Available at: <https://www.theguardian.com/world/2023/dec/01/the-gospel-how-israel-uses-ai-to-select-bombing-targets>. (accessed 14.2.24).

D'Cunha, T.R., Samit, 2023. IHL and Information Operations during Armed Conflict [WWW Document]. Lieber Inst. West Point. URL <https://lieber.westpoint.edu/ihl-and-information-operations-during-armed-conflict/> (accessed 8.3.24).

Department of the Army, 2015. Protection of Civilians (No. ATP 3-07.6). Available at: https://armypubs.army.mil/epubs/DR_pubs/DR_a/pdf/web/atp3_07x6.pdf. (accessed 9.2.24).

DIHR, 2020a. Introduction to human rights impact assessment | The Danish Institute for Human Rights [WWW Document]. Dan. Inst. Hum. Rights. URL <https://www.humanrights.dk/tools/human-rights-impact-assessment-guidance-toolbox/introduction-human-rights-impact-assessment> (accessed 14.2.24).

DIHR, 2020b. Human Rights Impact Assessment: Guidance and Toolbox. The Danish Institute for Human Rights, Copenhagen. Available at: https://www.humanrights.dk/files/media/dokumenter/udgivelser/hria_toolbox_2020/eng/dihr_hria_guidance_and_toolbox_2020_eng.pdf. (accessed 14.2.24).

Docherty, B., 2015. Mind the Gap (No. 9781623132408). Human Rights Watch. Available at: <https://www.hrw.org/report/2015/04/09/mind-gap/lack-accountability-killer-robots>. (accessed 27.2.24).
Docherty, B., Garlasco, M.E., 2003. Off target: the conduct of the war and civilian casualties in Iraq. Human Rights Watch, New York, NY.

Ekelhof, M., 2018a. Lifting the Fog of Targeting: "Autonomous Weapons" and Human Control through the Lens of Military Targeting. Nav. War Coll. Rev. 71.

Ekelhof, M., 2018b. Autonomous weapons: Operationalizing meaningful human control. ICRC Humanity. Law Policy. URL <https://blogs.icrc.org/law-and-policy/2018/08/15/autonomous-weapons-operationalizing-meaningful-human-control/> (accessed 5.2.24).

Every Casualty Counts, 2020. Standards for Casualty Recording. Every Casualty Worldwide. Available at: <https://everycasualty.org/wp-content/uploads/2022/09/Standards-Plain-English.pdf>. (accessed 11.12.23).
Forrester, 2020. The Emergence Of Offensive AI: How Companies Are Protecting Themselves Against Malicious Applications Of AI, Forrester Opportunity Snapshot. Available at: <https://www.snowdropsolution.com/pdf/The%20Emergence%20Of%20Offensive%20AI.pdf>. (accessed 24.2.24).

Forster, M., 2022. Refugee protection in the artificial intelligence era: a test case for rights. Royal Institute of International Affairs. Available at: <https://chathamhouse.souttron.net/Portal/Public/en-GB/RecordView/Index/191194>. (accessed 18.12.23). <https://doi.org/10.55317/9781784135324>

Future of Life Institute, 2024. EU Artificial Intelligence Act | Up-to-date developments and analyses of the EU AI Act. URL <https://artificialintelligenceact.eu/> (accessed 8.3.24).

Gaetjens, D., Devitt, K., Shanahan, C., 2022. Case Study: A Method for Ethical AI in Defence Applied to an Envisioned Tactical Command and Control System (No. DSTG-TR-3847). AU Department of Defence, Canberra.

Gazi, T., 2020. Data to the rescue: how humanitarian aid NGOs should collect information based on the GDPR. J. Int. Humanit. Action 5, 9. <https://doi.org/10.1186/s41018-020-00078-0>

Gisel, L., Olejnik, L., 2019. The potential human cost of cyber operations (Report). ICRC, Geneva. Available at: <https://www.icrc.org/en/document/potential-human-cost-cyber-operations>. (accessed 11.12.23).

Götzmann, N., 2019. Introduction to the Handbook on Human Rights Impact Assessment: Principles, methods and approaches, in: Handbook on Human Rights Impact Assessment. Edward Elgar Publishing, pp. 2–30.

Gubrud, M., Altmann, J., 2013. Compliance Measures for an Autonomous Weapons Convention (Working Paper #2). International Committee for Robot Arms Control. Available at: https://icrac.net/wp-content/uploads/2018/04/Gubrud-Altman_Compliance-Measures-AWC_ICRAC-WP2.pdf. (accessed 13.3.24).

Herrero, O., Volpicelli, G., Roussi, A., 2024. Tech giants pledge to fight AI deepfakes ahead of elections [WWW Document]. POLITICO. URL <https://www.politico.eu/article/tech-accord-industry-munich-security-conference-deepfake-ai-election-content/> (accessed 15.2.24).

Hirsh, M., 2024. How AI Will Revolutionize Warfare. Foreign Policy. URL <https://foreignpolicy.com/2023/04/11/ai-arms-race-artificial-intelligence-chatgpt-military-technology/> (accessed 20.2.24).

Hodgson, C., 2024. World's biggest tech companies pledge to fight AI-created election 'deepfakes' [WWW Document]. URL <https://www.ft.com/content/8dcbc162-a0f5-47ce-a2ea-5fb25cb160c5> (accessed 10.3.24).

Hoffmann, M., Frase, H., 2023. Adding Structure to AI Harm: An Introduction to CSET's AI Harm Framework. CSET. Available at: <https://cset.georgetown.edu/publication/adding-structure-to-ai-harm/>. (accessed 24.2.24).

Hofstra, K., Minor, E., 2014. Losing Sight of the Human Cost: Casualty Recording and Remote Control Warfare. Every Casualty Programme.

IBC, 2003. Iraq Body Count [WWW Document]. IraqBodyCount.org. URL <https://www.iraqbodycount.org/> (accessed 8.3.24).

ICRC, 2023a. Reducing Civilian Harm in Urban Warfare: A Handbook for Armed Groups (Publication). International Committee of the Red Cross. Available at: <https://www.icrc.org/en/publication/4682-reducing-civilian-harm-urban-warfare-handbook-armed-groups>. (accessed 24.2.24).

ICRC, 2023b. What you need to know about artificial intelligence in armed conflict [WWW Document]. URL <https://www.icrc.org/en/document/what-you-need-know-about-artificial-intelligence-armed-conflict> (accessed 4.2.24).

ICRC, 2022. Annual Report 2022, Volume 1.

ICRC, 2021a. International Committee of the Red Cross (ICRC) position on autonomous weapon systems: ICRC position and background paper. Int. Rev. Red Cross 915. <https://doi.org/9781623132408>

ICRC, 2021b. ICRC position on autonomous weapon systems.

ICRC, 2021c. Avoiding civilian harm from military cyber operations during armed conflicts.

ICRC, 2020a. Artificial intelligence and machine learning in armed conflict: A human-centred approach. Int. Rev. Red Cross 102, 463–479. <https://doi.org/10.1017/S1816383120000454>

ICRC, 2020b. Autonomous Weapon Systems: Implications of Increasing Autonomy in the Critical Functions of Weapons. International Committee of the Red Cross, Versoix. Available at: <https://shop.icrc.org/autonomous-weapon-systems-implications-of-increasing-autonomy-in-the-critical-functions-of-weapons.html>. (accessed 2.3.24).

ICRC, 2019. Applying principles of proportionality and precautions in Iraq: 2003. IHL Action. URL <https://ihl-in-action.icrc.org/case-study/iraq-computer-modelling-collateral-damage-estimates-and-choice-weapons> (accessed 24.2.24).

ICRC, 2018. A legal perspective: Autonomous weapon systems under international humanitarian law, in: Perspectives on Lethal Autonomous Weapon Systems, UNODA Occasional Papers. United Nations Office for Disarmament Affairs.

ICRC, 2015. Background report on international humanitarian law and the challenges of contemporary armed conflict (No. 32IC/15/11). Geneva. Available at: <https://rcrcconference.org/app/uploads/2015/10/32IC-Report-on-IHL-and-challenges-of-armed-conflicts.pdf>. (accessed 25.2.24).

ICRC, ARES, 2016. Explosive weapons in populated areas: Technical considerations relevant to their use and effects (Report). Armament Research Services and International Committee of the Red Cross, Perth. Available at: <https://www.icrc.org/en/document/explosive-weapons-populated-areas-use-effects>. (accessed 25.2.24).

Interviewee A (2024). Anonymised interview by Mariam Seklaou and Mykhailo Hohol, 23 February 2024

Interviewee B (2024). Anonymised interview by Madeleine Northfield and Mykhailo Hohol, 5 February 2024.

Interviewee C (2024). Anonymised interview by Dawson Shinnars and Mykhailo Hohol, 22 February 2024.

Interviewee D (2024). Anonymised interview by Madeleine Northfield and Mykhailo Hohol, 16 February 2024.

Interviewee E (2024). Anonymised interview by Mariam Seklaou and Mykhailo Hohol, 14 February 2024.

Interviewee F (2024). Anonymised interview by Dawson Shinnars and Mykhailo Hohol, 8 February 2024

Interviewee G (2024). Anonymised interview by Dawson Shinnars and Leo Podov Halbert, 20 February 2024.

Interviewee H (2024). Anonymised interview by Madeleine Northfield and Leo Podov Halbert, 6 February 2024.

Interviewee I (2024). Anonymised interview by Alex Pymm and Leo Podov Halbert, 14 February 2024.

Interviewee J (2024). Anonymised interview by Mariam Seklaou and Leo Podov Halbert, 2 February 2024.

iPRAW, 2021. Building Blocks for a Regulation of LAWS and Human Control – Updated Recommendations to the GGE (2021). International Panel on the Regulation of Autonomous Weapons. Available at: <https://web.archive.org/web/20221014074429/https://www.ipraw.org/publications/building-blocks/>. (accessed 23.2.24).

iPRAW, 2019. Verifying LAWS Regulation - Opportunities and Challenges. International Panel on the Regulation of Autonomous Weapons. Available at: <https://web.archive.org/web/20221104205545/https://www.ipraw.org/publications/working-paper-verification/>. (accessed 23.2.24).

iPRAW, 2018. Focus on the Human-Machine Relation in LAWS, “Focus on” Report No. 3. International Panel on the Regulation of Autonomous Weapons, Berlin.

ISD, 2022. Deadly Disinformation: how online conspiracies about Syria cause real world harm. The Syria Campaign. Available at: https://deadlydisinformation.org/source/Deadly_Disinformation_EN.pdf. (accessed 11.12.23).

Kimani, M., 2007. RTLM: the Medium that Became a Tool for Mass Murder, in: Thompson, A. (Ed.), The Media and the Rwanda Genocide. IDRC, pp. 110–124.

Klepper, D., 2023. Fake babies, real horror: False AI-generated images of the war in Gaza spark alarm [WWW Document]. Los Angel. Times. URL <https://www.latimes.com/world-nation/story/2023-11-29/israel-hamas-war-artificial-intelligence-deepfakes-disinformation> (accessed 7.2.24).

Kuner, C., Marelli, M., 2017. Handbook on Data Protection in Humanitarian Action.

Maathuis, C., Pieters, W., van den Berg, J., 2021. Decision support model for effects estimation and proportionality assessment for targeting in cyber operations. *Def. Technol.* 17, 352–374. <https://doi.org/10.1016/j.dt.2020.04.007>

Macnamara, J., 2005. *Media Content Analysis: Its Uses, Benefits and Best Practice Methodology*. Asia-Pac. Public Relat. J. 6.

Macnamara, J., 2003. *Mass Media Effects: A Review of 50 Years of Media Effects Research*.

Mantello, P., Ho, M.-T., 2023. Losing the information war to adversarial AI. *AI Soc.* s00146-023-01674-5. <https://doi.org/10.1007/s00146-023-01674-5>

Marcellino, W., Beauchamp-Mustafaga, N., Kerrigan, A., Chao, L.N., Smith, J., 2023. *The Rise of Generative AI and the Coming Era of Social Media Manipulation 3.0: Next-Generation Chinese Astroturfing and Coping with Ubiquitous AI*. RAND Corporation.

Marijan, B., 2022. *Autonomous Weapons: The False Promise of Civilian Protection* [WWW Document]. *Cent. Int. Gov. Innov.* URL <https://www.cigionline.org/articles/autonomous-weapons-the-false-promise-of-civilian-protection/> (accessed 13.2.24).

Meta, 2021. *Human Rights Due Diligence of Meta's Impacts in Israel and Palestine*. Meta. Available at: https://www.bsr.org/reports/BSR_Meta_Human_Rights_Israel_Palestine_English.pdf. (accessed 21.2.24).

Meta, 2020. *Here's how we're using AI to help detect misinformation* [WWW Document]. *ML Appl. Integr.* URL <https://ai.meta.com/blog/heres-how-were-using-ai-to-help-detect-misinformation/> (accessed 20.12.23).

Meta, 2018. *An Independent Assessment of the Human Rights Impact of Facebook in Myanmar*. Meta Newsroom. URL <https://about.fb.com/news/2018/11/myanmar-hria/> (accessed 25.2.24).

Morgan, F.E., Boudreaux, B., Lohn, A.J., Ashby, M., Curriden, C., Klima, K., Grossman, D., 2020. *Military Applications of Artificial Intelligence: Ethical Concerns in an Uncertain World*. RAND Corporation. Available at: https://www.rand.org/pubs/research_reports/RR3139-1.html. (accessed 21.1.24).

Moss, E., Watkins, E.A., Singh, R., Elish, M.C., Metcalf, J., 2021. *Assembling Accountability: Algorithmic Impact Assessment for the Public Interest*. *Data & Society*. Available at: <https://datasociety.net/library/assembling-accountability-algorithmic-impact-assessment-for-the-public-interest/>. (accessed 17.12.23).

NATO, 2021. *Allied Joint Doctrine for Joint Targeting Edition B Version 1* (No. AJP-3.9).

NATO, 2020. *The Protection of Civilians Allied Command Operations Handbook*. *Protection of Civilians Allied Command Operations*. Available at: <https://shape.nato.int/news-archive/2021/the-protection-of-civilians-allied-command-operations-handbook.aspx>. (accessed 23.1.24).

OHCHR, 2023. *Taxonomy of Human Rights Risks Connected to Generative AI*. United Nations Human Rights Office of the High Commissioner, Geneva. Available at: <https://www.ohchr.org/sites/default/files/documents/issues/business/b-tech/taxonomy-GenAI-Human-Rights-Harms.pdf>. (accessed 21.2.24).

OHCHR, 2019. *Guidance on Casualty Recording*. Office of the United Nations High Commissioner for Human Rights (OHCHR). Available at: https://www.ohchr.org/sites/default/files/Documents/Publications/Guidance_on_Casualty_Recording.pdf. (accessed 11.12.23).

OHCHR, 2011. *Guiding Principles On Business and Human Rights*. United Nations Human Rights Office of the High Commissioner, Geneva. Available at: https://www.ohchr.org/sites/default/files/documents/publications/guidingprinciplesbusinessshr_en.pdf. (accessed 25.2.24).

Pineiro, J.D., 2020. Gaining a Cognitive Advantage: Artificial Intelligence (AI) as a Decision Support System (DSS).

Rizk, J., Cordey, S., 2023. What we don't understand about digital risks in armed conflict and what to do about it [WWW Document]. Humanit. Law Policy Blog. URL <https://blogs.icrc.org/law-and-policy/2023/07/27/digital-risks-in-armed-conflict/> (accessed 8.3.24).

Roff, H., Moyes, R., 2016a. Meaningful Human Control, Artificial Intelligence and Autonomous Weapons. Presented at the Informal Meeting of Experts on Lethal Autonomous Weapons Systems, UN Convention on Certain Conventional Weapons, April 2016, Article36, Geneva.

Roff, H., Moyes, R., 2016b. Dataset: Artificial Intelligence, Autonomous Weapons, and Meaningful Human Control.

Romanosky, S., Goldman, Z., 2017. Understanding Cyber Collateral Damage. J. Natl. Secur. Law Policy 9.

Romanosky, S., Goldman, Z., 2016. What Is Cyber Collateral Damage? And Why Does It Matter? [WWW Document]. Lawfare. URL <https://www.lawfaremedia.org/article/what-cyber-collateral-damage-and-why-does-it-matter> (accessed 12.2.24).

Saleiro, P., Kuester, B., Hinkson, L., London, J., Stevens, A., Anisfeld, A., Rodolfa, K.T., Ghani, R., 2019. Aequitas: A Bias and Fairness Audit Toolkit.

Satter, R., 2024. Microsoft says it caught hackers from China, Russia and Iran using its AI tools. Reuters. Available at: <https://www.reuters.com/technology/cybersecurity/microsoft-says-it-caught-hackers-china-russia-iran-using-its-ai-tools-2024-02-14/>. (accessed 8.3.24).

Schmitt, M., 2017. Tallinn Manual 2.0 on the International Law Applicable to Cyber Operations, 2nd ed. Cambridge University Press, Cambridge. <https://doi.org/10.1017/9781316822524>

Shami, E., 2021. Assessing the risks of civilian harm from military cyber operations during armed conflicts. ICRC Humanit. Law Policy Blog. URL <https://blogs.icrc.org/law-and-policy/2021/06/22/risks-civilian-harm-cyber-operations/> (accessed 19.2.24).

Shushan, Y.B., 2024. "Netanyahu is the target": Microsoft report reveals Iran's cyber war on Israel [WWW Document]. Jerus. Post JPostcom. URL <https://www.jpost.com/breaking-news/article-785566> (accessed 8.3.24).

Simprints, 2023. A Responsible Biometric Deployment Handbook.

Srinivasan, S., 2024. Detecting AI fingerprints: A guide to watermarking and beyond [WWW Document]. Brookings. URL <https://www.brookings.edu/articles/detecting-ai-fingerprints-a-guide-to-watermarking-and-beyond/> (accessed 15.2.24).

Stewart, R., Hinds, G., 2023. Algorithms of war: The use of artificial intelligence in decision making in armed conflict [WWW Document]. ICRC Humanit. Law Policy Blog. URL <https://blogs.icrc.org/law-and-policy/2023/10/24/algorithms-of-war-use-of-artificial-intelligence-decision-making-armed-conflict/> (accessed 7.2.24).

Taddeo, M., Blanchard, A., 2022. A Comparative Analysis of the Definitions of Autonomous Weapons Systems. Sci. Eng. Ethics 28, 37. <https://doi.org/10.1007/s11948-022-00392-3>

Tavana, M., Trevisani, D.A., Kennedy, D.T., 2014. A Fuzzy Cyber-Risk Analysis Model for Assessing Attacks on the Availability and Integrity of the Military Command and Control Systems. Int. J. Bus. Anal. IJBAN 1, 21–36. <https://doi.org/10.4018/ijban.2014070102>

The Guardian, 2024. North Korea and Iran using AI for hacking, Microsoft says. The Guardian. Available at: <https://www.theguardian.com/technology/2024/feb/14/north-korea-iran-ai-hacking-microsoft>. (accessed 8.3.24).

Twomey, J., Ching, D., Aylett, M.P., Quayle, M., Linehan, C., Murphy, G., 2023. Do deepfake videos undermine our epistemic trust? A thematic analysis of tweets that discuss deepfakes in the Russian invasion of Ukraine. PLOS ONE 18, e0291668. <https://doi.org/10.1371/journal.pone.0291668>

UK Government, 2018. Human Machine Touchpoints: The United Kingdom's perspective on human control over weapon development and targeting cycles, in: Second Session. Presented at the CCW Group of Governmental Experts on lethal autonomous weapon systems, Geneva.

UK House of Lords, 2023. Proceed with Caution: Artificial Intelligence in Weapon Systems (No. HL Report 16).

UN Global Pulse, 2018. Experimenting with Big Data and Artificial Intelligence to Support Peace and Security. UN Global Pulse. Available at: <https://www.unglobalpulse.org/document/experimenting-with-big-data-and-ai-to-support-peace-and-security/>. (accessed 13.3.24).

UN OCHA, 2017. Compilation of military policy and practice: Reducing the humanitarian impact of the use of explosive weapons in populated areas, OCHA Policy and Studies Series. Available at: <https://reliefweb.int/report/world/compilation-military-policy-and-practice-reducing-humanitarian-impact-use-explosive>. (accessed 25.2.24).

UNICEF, UNHCR, 2023. Responsible Disaggregation of Data on Refugee and Other Forcibly Displaced Children: UNICEF-UNHCR Guidance Note.

US Joint Chiefs of Staff, 2009. No-Strike and the Collateral Damage Estimation Methodology (No. CJCSI 3160.01). US DoD. Available at: https://www.aclu.org/wp-content/uploads/legal-documents/drone_dod_3160_01.pdf. (accessed 19.2.24).

Watts, T., Bode, I., 2021a. Meaning-less human control: Lessons from air defence systems for lethal autonomous weapons. <https://doi.org/10.5281/ZENODO.4485695>

Watts, T., Bode, I., 2021b. Automation and Autonomy in Air Defence Systems Catalogue. <https://doi.org/10.5281/zenodo.4485694>

Woodcock, 2023. [New publication] The influence of AI decision-support systems on legal assessments in military targeting [WWW Document]. URL <https://www.asser.nl/about-the-asser-institute/news/new-publication-the-influence-of-ai-decision-support-systems-on-legal-assessments-in-military-targeting/> (accessed 8.3.24).

Wright, J., 2012. 'Excessive' ambiguity: analysing and refining the proportionality standard. Int. Rev. Red Cross 94, 819–854. <https://doi.org/10.1017/S1816383113000143>

Yamin, M.M., Ullah, M., Ullah, H., Katt, B., 2021. Weaponized AI for cyber attacks. J. Inf. Secur. Appl. 57, 102722. <https://doi.org/10.1016/j.jisa.2020.102722>

Zheng, C., Han, L., Ye, J., Zou, M., Liu, Q., 2009. A fuzzy comprehensive evaluation model for harms of computer virus. Presented at the 2009 IEEE 6th International Conference on Mobile Adhoc and Sensor Systems, MASS '09, pp. 708–713. <https://doi.org/10.1109/MOBHOC.2009.5336930>

Zoï Environment Network, 2022. Ecodozor [WWW Document]. Ecodozor.org. URL <https://ecodozor.org/index.php?lang=en> (accessed 10.3.24).

Zouave, E., Gustafsson, T., Bruce, M., Colde, K., 2020. Artificially Intelligent Cyberattacks (No. FOI-R-4947—SE). Totalförsvarets forskningsinstitut FOI. Available at: <https://www.foi.se/rapportsammanfattning?reportNo=FOI-R--4947--SE>.